

Week 9: Backtesting, Risk Analysis, and Taker Strategies

Assembling the full system – Milestone 3 (with a Coincall guest session)

North Carolina State University | Summer 2026

NC STATE UNIVERSITY

Industry Mentor: Dendi Suhubdy (Bitwyre) · Guest: Fenni Kang (Coincall)

Learning Objectives

- ▶ Build an event-driven backtest for options market making
- ▶ Produce a full P&L attribution and risk metrics
- ▶ Understand taker strategies and order-flow toxicity

Lecture 1: Backtest Architecture & Risk

- ▶ Event-driven simulation with deterministic replay
- ▶ The fill simulator; state tracking of inventory and Greeks
- ▶ Pitfalls: look-ahead bias, unrealistic fills, expiry handling
- ▶ Sharpe/Calmar, drawdown, block-bootstrap confidence intervals

Lecture 2 (Guest: Fenni Kang, Coincall): Taker Strategies

- ▶ When and why takers cross the spread
- ▶ Signals behind aggressive flow: momentum, liquidations, news
- ▶ Execution styles: sweep, TWAP/VWAP, iceberg-taking
- ▶ What taker behaviour means for the maker's order-flow toxicity

Milestone 3 Deliverable

- ▶ Full multi-Greek backtest on 3 months of BTC & ETH options
- ▶ P&L attribution sums to total within 1 bp; Sharpe with CI
- ▶ Greek exposures bounded; drawdown forensic + taker analysis
- ▶ Reproducible from config; report ≤ 20 pages